

L11: Robustness of Networks to Attacks

What happens in the case of attacks to the higher-degree nodes? What is the critical threshold f_c in that case?

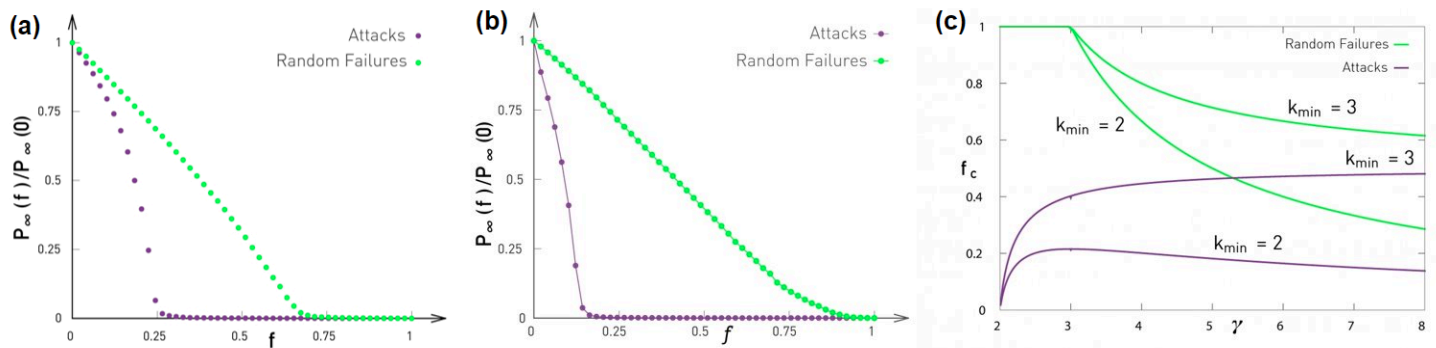


Image Source: **Network Science** by Albert-László Barabási .

Let us start with some simulations. Figure (a) contrasts the case of random failures and attacks in a $G(n,p)$ network with $n=10,000$ nodes and average degree $\bar{k} = 3$. The y-axis shows, as in the previous page, the probability that a node belongs in the largest connected component of the graph after we remove a fraction f of the nodes, normalized by the same probability when $f=0$. In the case of random failures, the critical threshold f_c is about 0.65-0.70 -- the theoretical prediction for an infinitely large network is $1 - 1/\bar{k} = 2/3$.

The same plot shows what happens when we “attack” the highest-degree nodes of the network. Specifically, we remove nodes iteratively, each time removing the node with the largest remaining degree, until we have removed a fraction f of all nodes.

This “attack” scenario destroys the LCC of the network for an even lower value of f than random removals. The critical threshold for attacks is about 0.25 in this network. The fact that the critical threshold for attacks is lower than for random failures should be expected – removing nodes with a higher degree makes the LCC sparser, increasing the chances that it will be broken into smaller components.

As we know from Lesson-4 however, $G(n,p)$ networks do not have hubs and the degree distribution is narrowly concentrated around the mean (Poisson distribution). *What happens in networks that have hubs – and in particular what happens in power-law networks in which the degree variance diverges?*

Figure (b) contrasts random failures and attacks for a power-law network $n=10,000$ nodes, degree exponent $\gamma = 2.5$, and minimum degree $k_{min} = 1$. Even though the case of random failures does not have a critical threshold (f_c tends to 1 as the network size increases), the case of attacks has a critical threshold that is actually even lower (around 0.2 in this example) than the corresponding threshold for $G(n,p)$ networks. Power-law networks have hubs, and attacks remove the hubs first. So, as the network's hubs are removed, the giant component quickly gets broken into many small components.

The moral of the story is that **even though power-law networks are robust to random failures, they are very fragile to attacks on their higher degree nodes.**

The mathematical analysis of this attack process is much harder (you can find it in your textbook in “Advanced topics 8.F”) and it does not lead to a closed-form solution for the critical threshold f_c .

Specifically, if the network has a power-law degree distribution with exponent γ and with minimum degree k_{min} , then the critical threshold f_c for attacks is given by the numerical solution of the equation:

$$f_c^{\frac{2-\gamma}{1-\gamma}} = 2 + \frac{2-\gamma}{3-\gamma} k_{min} \left(f_c^{\frac{3-\gamma}{1-\gamma}} - 1 \right)$$

Figure (c) shows numerical results for the critical threshold as a function of the degree exponent γ , for two values of k_{min} , and separately for failures and attacks.

The key points from this plot are:

- In the case of random failures, f_c decreases with γ – this is expected because as γ decreases towards 3, the degree variance increases, creating hubs with higher-degree.
- In the case of attacks, however, the relation between f_c and γ is more complex, and it also depends on k_{min} . In the case of $k_{min}=3$, as γ decreases, f_c decreases because the degree variance increases, the hubs have an even greater degree, and removing them causes more damage in the network's giant component.
- As expected, for given values of γ and k_{min} , attacks always have a smaller f_c than random failures.
- As γ increases, and for a given value of k_{min} , the critical threshold f_c for attacks approaches the critical threshold for random failures. The reason is that as γ increases, the variance of the degree distribution decreases. In the limit, the degree variance becomes zero and all nodes have the same degree. In that case, it does not matter if we remove nodes randomly or through attacks.

Food For Thought

How would you explain the way that the critical threshold f_c depends on the minimum degree k_{min} , for a given value of the exponent γ ? Answer separately for random failures and attacks.